

LEVEL 1: EASY (Basic Radical Simplification)

Exercise 1

Simplify the radical into mixed form:

$$\sqrt{72}$$

Answer 1

1. Prime factorization:

$$72 = 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3$$

2. Group factors into pairs for index $n = 2$:

$$\sqrt{72} = \sqrt{(2 \cdot 2) \cdot (3 \cdot 3) \cdot 2}$$

3. Extract grouped terms outside the radical:

$$= 2 \cdot 3\sqrt{2}$$

$$= \mathbf{6\sqrt{2}}$$

Exercise 2

Simplify the radical into mixed form:

$$\sqrt[3]{108}$$

Answer 2

1. Prime factorization:

$$108 = 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3$$

2. Group factors into triplets for index $n = 3$:

$$\sqrt[3]{108} = \sqrt[3]{(3 \cdot 3 \cdot 3) \cdot (2 \cdot 2)}$$

3. Extract grouped terms outside the radical:

$$= 3\sqrt[3]{2 \cdot 2}$$

$$= \mathbf{3\sqrt[3]{4}}$$

Exercise 3

Convert the expression into an entire radical:

$$4\sqrt{5}$$

Answer 3

1. We move the coefficient inside the radical

$$4\sqrt{5} = \sqrt{4^2 \cdot 5}$$

2. Multiply all factors inside the radical:

$$= \sqrt{16 \times 5}$$

$$= \mathbf{\sqrt{80}}$$

LEVEL 2: INTERMEDIATE (Radical Addition, Subtraction & Conversions)

Exercise 4

Simplify the radicals into mixed form and combine like terms:

$$\sqrt{75} - \sqrt{27} + \sqrt{48}$$

Answer 4

1. Factorize into prime factors and group into pairs ($n = 2$):

$$\sqrt{75} = \sqrt{(5 \cdot 5) \cdot 3} = 5\sqrt{3}$$

$$\sqrt{27} = \sqrt{(3 \cdot 3) \cdot 3} = 3\sqrt{3}$$

$$\sqrt{48} = \sqrt{(2 \cdot 2) \cdot (2 \cdot 2) \cdot 3} = 2 \cdot 2\sqrt{3} = 4\sqrt{3}$$

2. Combine like terms:

$$(5 - 3 + 4)\sqrt{3}$$

$$= 6\sqrt{3}$$

Exercise 5

Convert the expression into an entire radical:

$$2^3\sqrt{9}$$

Answer 5

1. Raise external factor to power of index $n = 3$, and move it inside the radical:

$$2^3\sqrt{9} = \sqrt[3]{2^3 \cdot 9}$$

2. Multiply all factors inside the radical:

$$= \sqrt[3]{8 \cdot 9}$$

$$= \sqrt[3]{72}$$

Exercise 6

Simplify the radical into mixed form:

$$\sqrt[4]{162}$$

Answer 6

1. Prime factorization:

$$162 = 2 \cdot 3 \cdot 3 \cdot 3 \cdot 3$$

2. Group factors into quadruplets for index $n = 4$:

$$\sqrt[4]{162} = \sqrt[4]{(3 \cdot 3 \cdot 3 \cdot 3) \cdot 2}$$

3. Extract grouped terms outside the radical:

$$= 3\sqrt[4]{2}$$

$$= 3\sqrt[4]{2}$$

Exercise 7

Simplify the radicals into mixed form and combine like terms:

$$\sqrt[3]{250} - \sqrt[3]{54}$$

Answer 7

1. Factorize into prime factors and group into triplets ($n = 3$):

$$\sqrt[3]{250} = \sqrt[3]{(5 \cdot 5 \cdot 5) \cdot 2} = 5\sqrt[3]{2}$$

$$\sqrt[3]{54} = \sqrt[3]{(3 \cdot 3 \cdot 3) \cdot 2} = 3\sqrt[3]{2}$$

2. Combine like terms:

$$(5 - 3)\sqrt[3]{2}$$

$$= 2\sqrt[3]{2}$$

LEVEL 3: HARD (Algebraic Radicals & Higher Order Operations)

Exercise 8

Convert the expression into an entire radical:

$$3x^2y\sqrt[3]{2x}$$

Answer 8

1. Raise external factors to power of index $n = 3$ and move them inside the radical:

$$3x^2y\sqrt[3]{2x} = \sqrt[3]{27 \cdot x^6 \cdot y^3 \cdot 2x}$$

2. Combine all factors inside the radical:

$$\sqrt[3]{27 \cdot x^6 \cdot y^3 \cdot 2x} = \sqrt[3]{54x^7y^3}$$

$$= \sqrt[3]{54x^7y^3}$$

Exercise 9

Simplify the radicals into mixed form and combine like terms ($x \geq 0$):

$$\sqrt{72x^5} - x\sqrt{18x^3} + \sqrt{50x^5}$$

Answer 9

1. Factorize numerical and variable terms into prime components, grouping into pairs ($n = 2$):

$$\begin{aligned} 1. \quad \sqrt{72x^5} &= \sqrt{(2 \cdot 2) \cdot (3 \cdot 3) \cdot (x \cdot x) \cdot (x \cdot x) \cdot 2 \cdot x} \\ &= 2 \cdot 3 \cdot x \cdot x \sqrt{2x} = 6x^2\sqrt{2x} \end{aligned}$$

$$\begin{aligned} 2. \quad x\sqrt{18x^3} &= x\sqrt{(3 \cdot 3) \cdot (x \cdot x) \cdot 2 \cdot x} \\ &= x \cdot (3 \cdot x) \sqrt{2x} = 3x^2\sqrt{2x} \end{aligned}$$

$$\begin{aligned} 3. \quad \sqrt{50x^5} &= \sqrt{(5 \cdot 5) \cdot (x \cdot x) \cdot (x \cdot x) \cdot 2 \cdot x} \\ &= 5 \cdot x \cdot x \sqrt{2x} = 5x^2\sqrt{2x} \end{aligned}$$

2. Combine like terms:

$$(6x^2 - 3x^2 + 5x^2)\sqrt{2x}$$

$$= 8x^2\sqrt{2x}$$

Exercise 10

Simplify the radicals into mixed form and combine like terms ($x, y \geq 0$):

$$2y \sqrt[4]{48x^9y^4} - x^2 \sqrt[4]{243xy^8}$$

Answer 10

1. Factorize numerical and variable terms into prime components, grouping into quadruplets ($n = 4$):

$$\begin{aligned} 1. \quad & 2y \sqrt[4]{48x^9y^4} = \\ & 2y \sqrt[4]{(2^4) \cdot (x^4) \cdot (x^4) \cdot (y^4) \cdot 3 \cdot x} = \\ & = 2y \cdot (2 \cdot x^2 \cdot y) \sqrt[4]{3x} = 4x^2y^2 \sqrt[4]{3x} \end{aligned}$$

$$\begin{aligned} 2. \quad & x^2 \sqrt[4]{243xy^8} = \\ & x^2 \sqrt[4]{(3^4) \cdot (y^4) \cdot (y^4) \cdot 3 \cdot x} = \\ & = x^2 \cdot (3 \cdot y^2) \sqrt[4]{3x} = 3x^2y^2 \sqrt[4]{3x} \end{aligned}$$

2. Combine like terms:

$$(4x^2y^2 - 3x^2y^2) \sqrt[4]{3x}$$

$$= x^2y^2 \sqrt[4]{3x}$$